

Sri Lanka Institute of Information Technology



IE2102 - Network Programming Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 02

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```
ubuntu@ubuntu: ~/nplab2
GNU nano 7.2 lab2_py_udp_server.py
import socket
import argparse

HOST = '0.0.0.0'
def server(port):
    sock = socket.socket(socket.AF_INET , socket.SOCK_DGRAM)
    sock.bind((HOST , port))
    print("UDP server listening on port ",port)

    while True:
        data , addr = sock.recvfrom(4096)
        if not data :
            continue

        message = data.decode(errors='ignore')
        print("from" , addr , ":" , message)

        # send ACK back to client
        sock.sendto(b'ACK' , addr)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--port" , type=int , required=True)
    args = parser.parse_args()

    server(args.port)
```

```
ubuntu@ubuntu: ~/nplab2
GNU nano 7.2 lab2_c_udp_sender.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <sys/time.h>
#include <sys/select.h>

#define MAX_MSG 1000
#define RETRY_LIMIT 3

int main(int argc , char *argv[]){
    if(argc != 3){
        printf("Usage : %s <server_ip> <server_port>\n" , argv[0]);
        return 1;
    }
    char *server_ip = argv[1];
    int server_port = atoi(argv[2]);

    // create the socket
    int s = socket(AF_INET , SOCK_DGRAM , 0);
    if(s < 0){
        perror("socket creation failed");
        return 1;
    }
}
```

[Read 89 lines]

```

        perror("socket creation failed");
        return 1;
    }

    struct sockaddr_in server_addr;
    memset(&server_addr , 0 , sizeof(server_addr));
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(server_port);
    inet_pton(AF_INET , server_ip , &server_addr.sin_addr);

    int total = 0, acked = 0 , lost = 0;

    printf("sending %d UDP message to %s:%d\n" , MAX_MSG , server_ip , server_port);

    for(int i =0; i < MAX_MSG; i++){
        char msg[32];
        snprintf(msg , sizeof(msg) , "Message-%d" , i);

        int got_ack = 0;

        for(int retry = 0; retry < RETRY_LIMIT; retry++){
            sendto(s , msg , strlen(msg) , 0 ,(struct sockaddr*)&server_addr , sizeof(server_addr));
            total++;

            fd_set fds;
            FD_ZERO(&fds);
            FD_SET(s,&fds);

```

```

            FD_ZERO(&fds);
            FD_SET(s,&fds);

            struct timeval tv;
            tv.tv_sec = 0;
            tv.tv_usec =20000;

            int rv = select(s+1 , &fds , NULL , NULL , &tv);
            if(rv > 0){
                char buf[64];
                int n = recvfrom(s , buf , sizeof(buf)-1 , 0, NULL , NULL);
                if (n > 0){
                    buf[n] = 0;
                    got_ack = 1;
                    acked++;
                    break;
                }
            }
        }

        if(!got_ack){
            lost++;
        }

        if((i+1) % 100 == 0){
            printf("sending %d message .... \n", i+1 );

```

```

    }

    if(!got_ack){
        lost++;
    }

    if((i+1) % 100 == 0){
        printf("sending %d message .... \n", i+1 );
    }
}

printf("\n====RESULT====\n");
printf("total send : %d\n" , total);
printf("ACK received :%d\n" , acked);
printf("packets lost :%d\n" , lost);
printf("loss percentage : %2f%%\n" , (lost * 100.0/total));
printf("=====\n");

close(s);
return 0;
}

```

```

ubuntu@ubuntu:~$ ls
act1    act3.c  act5    act6.c  Documents  nplab1  Public  Videos
act1.c  act4    act5.c  cd      Downloads  nplab2  snap
act3    act4.c  act6    Desktop Music      Pictures Templates
ubuntu@ubuntu:~$ cd nplab2
ubuntu@ubuntu:~/nplab2$ ls
client  lab2_c_udp_sender.c  lab2_py_udp_server.py
ubuntu@ubuntu:~/nplab2$ nano lab2_py_udp_server.py
ubuntu@ubuntu:~/nplab2$ nano lab2_c_udp_sender.c
ubuntu@ubuntu:~/nplab2$ python3 lab2_py_udp_server.py --port 5000
UDP server listening on port 5000

```

```
ubuntu@ubuntu: ~/nplab2
ubuntu@ubuntu: ~/nplab2
ubuntu@ubuntu:~/nplab2$ gcc lab2_c_udp_sender.c -o client
ubuntu@ubuntu:~/nplab2$ ./client 127.0.0.1 5000
sending 1000 UDP message to 127.0.0.1:5000
sending 100 message ....
sending 200 message ....
sending 300 message ....
sending 400 message ....
sending 500 message ....
sending 600 message ....
sending 700 message ....
sending 800 message ....
sending 900 message ....
sending 1000 message ....

=====RESULT=====
total send : 1000
ACK received :1000
packets lost :0
loss percentage : 0.000000%
=====
ubuntu@ubuntu:~/nplab2$
```

```
ubuntu@ubuntu:~/nplab2$ nano lab2_py_udp_server.py
ubuntu@ubuntu:~/nplab2$ nano lab2_c_udp_sender.c
ubuntu@ubuntu:~/nplab2$ python3 lab2_py_udp_server.py --port 5000
UDP server listening on port 5000
from ('127.0.0.1', 53890) : Message-0
from ('127.0.0.1', 53890) : Message-1
from ('127.0.0.1', 53890) : Message-2
from ('127.0.0.1', 53890) : Message-3
from ('127.0.0.1', 53890) : Message-4
from ('127.0.0.1', 53890) : Message-5
from ('127.0.0.1', 53890) : Message-6
from ('127.0.0.1', 53890) : Message-7
from ('127.0.0.1', 53890) : Message-8
from ('127.0.0.1', 53890) : Message-9
from ('127.0.0.1', 53890) : Message-10
from ('127.0.0.1', 53890) : Message-11
from ('127.0.0.1', 53890) : Message-12
from ('127.0.0.1', 53890) : Message-13
from ('127.0.0.1', 53890) : Message-14
from ('127.0.0.1', 53890) : Message-15
from ('127.0.0.1', 53890) : Message-16
from ('127.0.0.1', 53890) : Message-17
from ('127.0.0.1', 53890) : Message-18
from ('127.0.0.1', 53890) : Message-19
from ('127.0.0.1', 53890) : Message-20
from ('127.0.0.1', 53890) : Message-21
```

```
from ('127.0.0.1', 53890) : Message-973  
from ('127.0.0.1', 53890) : Message-974  
from ('127.0.0.1', 53890) : Message-975  
from ('127.0.0.1', 53890) : Message-976  
from ('127.0.0.1', 53890) : Message-977  
from ('127.0.0.1', 53890) : Message-978  
from ('127.0.0.1', 53890) : Message-979  
from ('127.0.0.1', 53890) : Message-980  
from ('127.0.0.1', 53890) : Message-981  
from ('127.0.0.1', 53890) : Message-982  
from ('127.0.0.1', 53890) : Message-983  
from ('127.0.0.1', 53890) : Message-984  
from ('127.0.0.1', 53890) : Message-985  
from ('127.0.0.1', 53890) : Message-986  
from ('127.0.0.1', 53890) : Message-987  
from ('127.0.0.1', 53890) : Message-988  
from ('127.0.0.1', 53890) : Message-989  
from ('127.0.0.1', 53890) : Message-990  
from ('127.0.0.1', 53890) : Message-991  
from ('127.0.0.1', 53890) : Message-992  
from ('127.0.0.1', 53890) : Message-993  
from ('127.0.0.1', 53890) : Message-994  
from ('127.0.0.1', 53890) : Message-995  
from ('127.0.0.1', 53890) : Message-996  
from ('127.0.0.1', 53890) : Message-997  
from ('127.0.0.1', 53890) : Message-998  
from ('127.0.0.1', 53890) : Message-999
```